FALL'ITIS

Maze Rescue Simulation

RoboCup, Bordeaux, France 2023

Our Team

Our team consists of two members from Italy, who are passionate about robotics and computer vision. Unfortunately, we experienced the departure of our third member, Lorenzo Corradin, due to an overwhelming workload**.** We express our gratitude to Lorenzo for his valuable contributions to the project. Additionally, we are immensely thankful to our mentor, Fabrizio Vendramin, for providing us with the opportunity to participate in this competition

From left to right: Fabrizio, Lorenzo, Michele, Alin

Alin Odobescu has been involved in developing the mapping system, path planning and following. Moreover, this competition has ignited his enthusiasm for machine learning and computer vision.

Michele Tesser has dedicated his efforts to object detection and dataset management, operate scoring game elements and analyzing various corner cases. His passion for mathematics has been greatly involved in the competition.

We are proud to have emerged victorious in the national phase of this competition. The opportunity to compete at such a high level is an immense honor for us, and we are deeply grateful for it. Throughout the competition, we will continue to showcase our skills and dedication.

Our Software

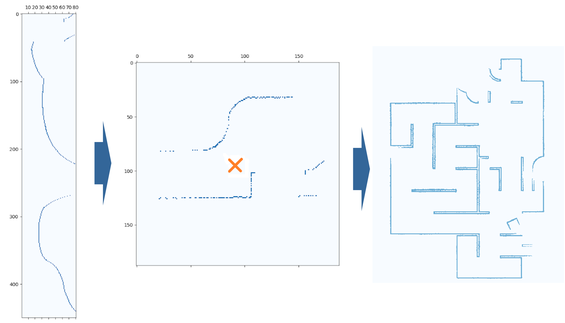
Our software stands out due to its strong independence from the tile system. We have made significant efforts to ensure that our code is adaptable and versatile, allowing it to function effectively regardless of the tile system.

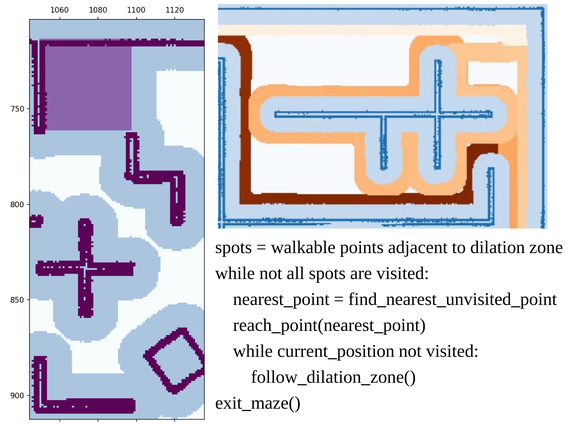
MAPPING

One of our key strengths lies in our mapping capabilities. We employ Lidar technology to construct a map of the maze. Our mapping solution utilizes a nonprobabilistic approach, utilizing a grid-based system to differentiate between empty spaces and walls. By converting Lidar data into a polar representation, we obtain distance and angle information for each point. This data is then transformed into a Cartesian grid and incorporated into the occupancy grid for a global mapping.

MAPPING BONUS

We have gone beyond the minimum requirements for mapping by employing a high sampling density. Through a down-sampling process, we analyze small chunks of the high-resolution grid, checking for wall hits in each area. This enables us to create a bonus map capable of meeting the requirements.





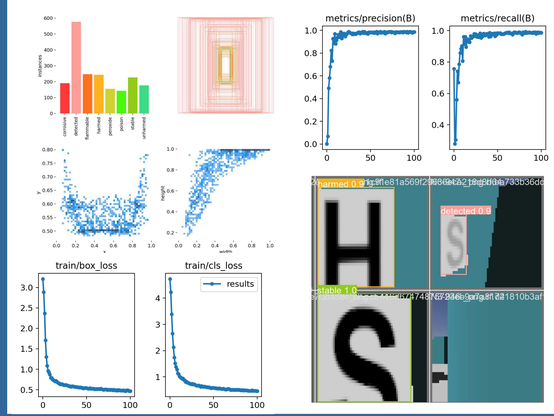
```
spots = walkable points adjacent to dilation zone
while not all spots are visited:
    nearest_point = find_nearest_unvisited_point
    reach_point(nearest_point)
    while current_position not visited:
        follow_dilation_zone()
exit_maze()
```

NAVIGATING THE MAZE

Navigating the maze has been a significant challenge. Our path and motion planning algorithm takes into account various inputs, including the mapping data, the robot's position, and yaw. It outputs the appropriate wheel speeds. Inside the algorithm, we have implemented a hidden layer where we store short and long planned paths. We rely on the hand rule and utilize breadth-first search to search for positions and execute small movements, overcoming the complexities of the maze.

SIGN DETECTION

We initially employed a convolutional neural network (CNN) for image labeling. However, we encountered limitations in locating signs, and in many corner cases. To address this, we made the difficult decision to switch to an object pose estimation real-time system. By combining object detection using the YOLO model (trained on a dataset of approximately 3,000 labeled photos) with rotation, scalar, and vector multiplication techniques, we were able to enhance our sign detection capabilities significantly. This approach provides more accurate and reliable identification of victims and hazmats within the maze.



LiDAR

One of the standout features of our robot is the LiDAR (Light Detection and Ranging) sensor. We chose LiDAR over other distance sensors due to its greater utility and power. With the ability to emit 2048 rays in 4 layers, covering a 360° fov, the LiDAR provides us with an environmental visualization. While there were cheaper distance sensors available, they would have limited our ability to effectively visualize the surroundings. The LiDAR's capability to emit multiple rays allows us to have more detailed understanding of the robot's environment.

Color Sensor

Situated on the robot's base, we have integrated a color sensor to detect the color of the tile the robot is positioned on. While considering computer vision as an alternative, we deemed it excessive for our needs. The color sensor emits a ray and returns a tensor that indicates the detected color.

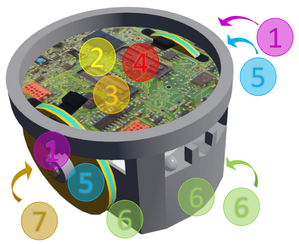
GPS

To precisely determine the robot's location, we have incorporated a GPS module. While odometry could have provided relative position information, certain scoring elements necessitate an absolute position. Therefore, we opted for GPS, which offers a vector with the robot's three coordinates.

Inertial Unit

Our robot features an inertial unit that goes beyond a gyroscope, as it includes mathematical functions for converting angular velocities into angles. The inertial unit is essential for determining the robot's orientation, allowing us to create a global map rather than just a local one. Odometry alone would not suffice since we need to account for the new unknown orientation after a Lack of Progress.

Our Robot



1. CAMERA
2. LiDAR
3. GPS
4. INERTIAL UNIT
5. WHEEL AND MOTOR
6. DISTANCE SENSOR
7. COLOR SENSOR

Distance Sensors

We use three distance sensors on the base of our robot for detecting black holes. While limited to using only one color sensor, it would not have been sufficient to detect black holes since the robot's movements are not confined to cardinal directions. These distance sensors measure the distance to the ground and enable us to identify unexpected increases that indicate the presence of a hole.

Cameras

Cameras are indispensable for our robot as they enable us to capture images of the maze and facilitate victim detection. We have chosen to integrate two cameras, positioned at the right and left edges, aligned with the center axis to simplify mathematical calculations.